PRODUCT REQUIREMENTS DOCUMENT

Amity Resume Builder

A free, mobile-first resume builder using a controlled university template, with optional job-description-based AI assistance.

Document field	Value
Version	1.0
Status	Draft for product, placement-cell and technical review
Prepared on	10 October 2026
Primary users	University students preparing placement and internship resumes
Initial platform	Responsive web app / installable PWA; mobile-first
Core promise	Create and download a clean, consistent resume for free from a phone

Important template dependency

The uploaded three-page resume is the visual reference available for this PRD, not a verified editable master template. Before production, the placement cell must approve the exact template version, required sections, logo asset, photo rules, and declaration wording. The app must not claim official university endorsement until authorization is obtained.

1. Executive Summary

Amity Resume Builder helps students create a placement-ready resume without needing a laptop or specialist formatting skills. A student fills in structured fields on a phone, adds a profile photo and professional profile links, sees a live preview in the approved university format, and downloads a PDF with functional hyperlinks.

The product has two creation modes. Manual Mode gives students direct control over every section. JD Assist lets a student paste or upload a job description and receive tailored suggestions based only on their own profile data. AI is an assistant, not an authority: it may reorganize, prioritize, and rewrite verified information, but it must not invent skills, credentials, work experience, projects, metrics, awards, or personal details.

The initial release should prioritize correctness, reproducible formatting, privacy, and low-friction mobile use over a large feature set. It should not promise ATS success or imply that an AI-generated resume guarantees an interview.

2. Problem Statement and Opportunity

Students often find resume creation difficult because they must write professional content, know what to include, format many sections consistently, add a photo, and create hyperlinks correctly. On a phone, editing a document template is cumbersome. A university format adds another constraint: students need their own details presented within a common structure.

Problem to solve

- Reduce the effort and formatting knowledge needed to prepare a resume for placement season.
- Keep every resume consistent with a placement-cell-approved university template.
- Make contact and profile links easy to enter while displaying familiar labels such as “GitHub” rather than long raw URLs.
- Help students tailor relevant wording to a job description without falsifying their profile.
- Make the essential workflow free and usable from a phone.

Product principles

- Accuracy before polish: never trade factual correctness for stronger-sounding content.
- Student control: AI suggestions are visible, editable, and explicitly accepted by the student.
- No blank-section clutter: omit optional empty sections; show required sections and explain what is missing.
- Predictable rendering: the template is versioned and locked; user content changes, template rules do not.
- Privacy by default: collect only what is needed, be clear about AI processing, and provide delete controls.
- Free core workflow: manual entry, preview, and final PDF export must not be paywalled in the MVP.

3. Goals, Non-Goals and Success Measures

Goals

Goal	Proposed success measure
Mobile usability	A typical student can complete a basic resume on a supported phone without desktop-only steps.
Reliable output	At least 99% successful PDF exports in release testing; no critical clipping or lost content in the approved test set.
Content fidelity	100% of manually entered facts appear accurately in the export; no silent truncation.
AI factual safety	Zero unsupported factual claims in the pre-release evaluation set; each generated claim has an internal source mapping.
Low friction	Median completion time under 10 minutes for a student with their information ready (validate with usability testing).
Accessible experience	Target WCAG 2.2 AA-informed mobile behavior, including keyboard access, visible focus, readable contrast, and alternatives to drag gestures.

Non-goals for MVP

- A general-purpose, multi-template resume marketplace. MVP launches with one approved Amity template version.

- Automatic web scraping of LinkedIn, GitHub, job portals, or student profiles.
- Automatically submitting job applications or sending resumes to recruiters.
- An ATS score guarantee, job-placement prediction, or claims that a resume will pass every screening system.
- AI inventing missing projects, certifications, skills, grades, dates, metrics, or employment history.
- A full word processor with arbitrary page design controls.

4. Users and Core User Stories

Persona	Need	Key user story
Student with little resume-writing experience	Guided form and a good default layout	As a student, I want clear questions for each section so I can fill the resume without learning document formatting.
Student working only on a phone	Responsive fields, image upload, preview, download	As a phone user, I want to finish and export my resume without needing a laptop.
Student applying to a specific JD	Relevant suggestions grounded in their profile	As an applicant, I want my existing experience and projects presented in the language of a job description without adding claims I cannot support.
Placement administrator / template owner	Controlled template updates and consistency	As a template owner, I want to approve template changes once so all students receive the same official structure.

5. Scope and Release Plan

MVP: Must ship

- Mobile-first responsive web app with a simple start screen: “Build manually” or “Tailor to a job description”.
- One versioned, placement-cell-approved university resume template; brand assets loaded from approved source files.
- Structured forms for personal details, academics, experience, projects, technical skills, achievements/certifications, leadership/initiative, profile links, and declaration details, subject to the approved schema.
- Photo upload with crop/position controls and clear file/type/size errors.
- Add, edit, reorder (where template permits), and remove repeatable experience/project/achievement entries.
- URL parsing and validation for GitHub, LinkedIn, LeetCode, portfolio and other allowed links; show a short label with a clickable target.
- Live preview, missing-required-field checklist, safe draft saving on the current device, and a clear “delete draft” action.
- PDF export with clickable hyperlinks, embedded photo/logo, selectable text, predictable pagination, and no silent truncation.
- Optional JD text entry or supported PDF/DOCX/TXT upload, extraction review, AI suggestions, source mapping, and human approval before use.
- Privacy notice and explicit consent before transmitting profile/JD content to an AI provider; safe upload handling and data minimization.

Later releases

- Optional account-based cloud save and cross-device resume management.
- Additional approved university templates or program-specific variants if authorized.
- DOCX export, duplicate resume versions, Hindi/English interface localization, and richer accessibility support after validation.
- Placement-cell template administration console with approval workflow and rollback.
- Optional parsing of an existing resume to prefill fields, with full source review and no overwrite without confirmation.

6. Primary User Journeys

Journey A: Manual resume

1. Open the app on a phone and select “Build manually”. No account is required to complete the core build.
2. See a short explanation of the approved template and privacy choices. AI processing is not used in manual mode.
3. Complete required fields with section-specific examples and validation. Optional sections can be skipped.
4. Add profile links by pasting the URL; the form previews the final label and destination.
5. Upload a portrait photo if required by the approved template; crop and preview it in the actual header position.

6. Review the live resume and a checklist of missing required data, invalid links, and potential overflow.
7. Download the PDF. Keep the draft on the current device only if the user elects to save it; offer a clear delete/reset option.

Journey B: JD Assist

1. Choose “Tailor to a job description” and paste JD text or upload a supported file within documented size/type limits.
2. The app extracts text and shows the extracted content for review before AI analysis; unreadable or incomplete files trigger a clear error and text-paste fallback.
3. The student fills or confirms their profile. JD Assist cannot produce factual candidate claims from the JD alone.
4. The system presents extracted role title, key responsibilities, required skills, preferred skills, and qualifications with confidence/ambiguity indicators.
5. The AI suggests edits for eligible sections and marks each suggestion as supported by profile data, needing confirmation, or unavailable from the profile.
6. The student accepts, edits, or rejects each suggestion. Existing manually entered facts are never overwritten silently.
7. The student reviews the final template preview and downloads the PDF. The JD and profile are not retained on the server beyond the declared processing/retention policy unless the user opts in.

7. Template and Content Model

Template governance

The sample PDF shows a header with Amity University logo, candidate photo, name and contact details; dark section bars; and sections including Career Objective, Academic Details, Experience, Projects, Technical Skills, Awards and Achievements, Leadership & Initiative, Links, and Self Declaration. Treat this as a reference list. The placement cell must confirm the required ordering, exact labels, compulsory fields, maximum pages, and approved assets before implementation.

Area / section	Candidate fields	Display rule
Header	Full name; phone; email; address/city if mandated; student photo; approved university logo	Use the locked header layout. Validate contact formats; never synthesize contact data. Photo requirement and address granularity must be approved.
Career Objective / Overview	Student-authored text, or AI suggestion based on the verified profile and selected JD	Keep manual text unchanged unless the student asks to edit it. AI output is a draft requiring approval.
Academic Details	Degree/program, institution, dates/year, CGPA/percentage, secondary/senior-secondary details if required, relevant minor/specialization	No inferred marks, grades, dates or credentials. Support clearly labeled grading systems.
Experience	Role, organization, dates, format/location if requested, bullet descriptions, optional evidence/context	Repeatable entries. Do not infer employer names, dates, seniority, metrics or outcomes.
Projects	Project name, technology stack, description/bullets, optional project URL	Repeatable entries. Technology list must reflect the candidate’s input; AI may reorder or rewrite but not add tools.
Technical Skills	Skills by category; proficiency only if user supplied/confirmed it	Display only populated categories. JD relevance may reorder verified skills but cannot assert skills absent from profile.
Awards / Achievements / Certifications	Title, issuer/event, date/year, optional proof URL	Optional repeatable entries. Do not convert participation into a win or assume a credential is certified.
Leadership & Initiative	Activity, role, contribution, dates, user-written details	Omit when blank if the approved template permits; no generic filler claims.
Links	Type/label and URL for GitHub, LinkedIn, LeetCode, portfolio or other allowed profiles	Display short labels as hyperlinks. Omit missing links entirely; never render “N/A” or placeholder URLs.

Area / section	Candidate fields	Display rule
Self Declaration / Footer	Approved declaration text; date; place; typed name or signature if required	Use the exact authorized wording; date/place optionality must follow approved template.

Section presence rules

- Mandatory sections are configured by the approved template and validated before export.
- Optional section with no valid content is omitted entirely; do not leave blank headings, empty bullets, “N/A”, or fake filler text.
- If a required section is empty, show a direct completion prompt and block final export only if the placement-cell-approved rule requires it. Otherwise warn clearly.
- Repeated entries follow student-selected order or the approved template order; the AI may recommend relevance ranking, but the student controls final order.

8. Hyperlink and Media Requirements

Profile URLs

- Input field accepts a full URL. The UI should explain where to copy the link from, and may offer a “paste profile URL” example without requiring OAuth.
- Normalize safe common forms (for example, add https:// only when an input is clearly a bare domain/path and can be parsed unambiguously). Show the normalized final destination before saving.
- Allow only approved schemes, normally HTTPS (HTTP may be rejected or upgraded only when the transformation is safe and shown to the user). Reject javascript:, data:, file:, malformed URLs, and unsupported schemes.
- Validate host/format, not account ownership. The app must not say a profile exists or belongs to the student unless independently verified with permission.
- Resume display text defaults to labels such as GitHub, LinkedIn, LeetCode, Portfolio, or a student-provided short label. The full URL remains the clickable target, not visible resume text.
- PDF export tests must confirm each displayed label points to the intended URL and is actually clickable in common PDF viewers.

Photo and logo

- Use only approved logo files supplied or authorized by the placement cell; do not extract a low-resolution logo from a sample output and treat it as a production asset.
- Support common image types such as JPEG and PNG, set a documented file size limit, validate file signatures and reject malformed files.
- Offer crop/position/zoom within the fixed frame. Never stretch or distort an image. Preview the final crop in context.
- Strip unnecessary image metadata from exported media where practical; do not store a photo remotely unless the retention notice clearly explains why and the student consents.

9. JD Assist: AI Behavior and Accuracy Controls

Supported inputs and outputs

Stage	Requirement
JD input	Accept pasted text and supported PDF/DOCX/TXT files. Apply size/page limits, malware-safe parsing, and extraction error reporting. No macros or executable content is processed.
JD extraction	Extract role title, responsibilities, required/preferred skills, qualifications, and keywords. Show extracted text/key points for user confirmation when confidence is low.
Candidate facts	Build a structured, source-traceable profile from the student’s own inputs and explicitly confirmed facts. A JD is evidence of employer requirements, not evidence that the student has a skill.
Suggestions	Propose wording for career overview/objective, project/experience bullets, skill ordering, and relevant section emphasis. Do not auto-create unsupported items.

Stage	Requirement
Review UI	Show original text, suggested version, changes/diff, supporting profile field(s), uncertainty notes, and Accept/Edit/Reject controls.
Final output	Only student-approved content is exported. Clearly separate confirmed facts from optional claims that still need student verification.

Hard rules for factual integrity

- Every generated factual statement must map internally to one or more candidate-supplied or user-confirmed fields. This source mapping need not appear as citations on the resume, but must be inspectable in the review UI or audit record.
- Never invent or infer employers, job titles, degrees, grades, dates, durations, certifications, project names, programming languages, frameworks, proficiency levels, metrics, team sizes, revenue, users, awards, or measurable outcomes.
- Do not turn exposure, coursework, or a JD keyword into demonstrated proficiency. If the JD asks for a missing skill, display “Not listed in your profile” and invite the student to add it only if true.
- Any number or quantified result must come from user input or be removed. Do not fabricate percentages or impact metrics to make bullets sound stronger.
- AI may shorten, reorder, or rephrase supported content. It must not change the meaning, level of ownership, scope, tense, or degree of certainty.
- Student-authored fields are immutable by default. Suggestions go to a review queue; accepting a suggestion is an explicit edit.
- The system should fail closed when evidence mapping fails: do not show the unsupported output as resume-ready content.

Different students applying to the same JD

The product should not generate the same generic objective for every applicant. Personalization should come from each student’s actual projects, skills, degree, internship, achievements, and stated experience, then prioritize the most relevant facts for the given JD. Wording can vary through controlled alternatives, but all alternatives must preserve the same evidence and meaning. If two students have identical verified profiles and the same JD, the system must not fabricate differences merely to appear unique. Do not promise a uniqueness guarantee; measure useful profile-specific variation and factual fidelity instead.

Similarity and diversity check

- Evaluate test cohorts with the same JD but different profiles. The accepted outputs should reference profile-specific supported details where relevant.
- Compare outputs for excessive boilerplate similarity, but do not force variation where evidence is identical.
- Provide two or more wording alternatives when a suggestion can be expressed differently without changing facts; keep alternatives within the same source constraints.
- Never use random skills, accomplishments or claims as a uniqueness mechanism.

Prompt injection and untrusted JD text

Treat all uploaded JD content as untrusted data, not as instructions to the AI system. Ignore any text that tells the model to reveal secrets, change system rules, fetch external data, or insert claims into the resume. The model should only analyze the JD as job-related content. Use structured prompts, bounded context, output schemas, and server-side validation; do not rely on prompt wording alone.

10. Functional Requirements

ID	Feature	Requirement	Priority
FR-01	Start mode	Offer Manual Build and JD Assist; explain the difference and that manual mode does not require AI.	Must
FR-02	Profile form	Collect approved fields in clear sections with inline help, required/optional state, and validation.	Must

ID	Feature	Requirement	Priority
FR-03	Repeatable lists	Add/edit/remove multiple academic, experience, project, skill category, certification and achievement entries as allowed.	Must
FR-04	Conditional fields	Ask only relevant follow-up fields; hide optional empty sections in the resume.	Must
FR-05	Photo upload	Upload, validate, crop and preview photo against the approved template frame.	Must
FR-06	Hyperlinks	Validate URL and render a short visible label with a clickable URL destination.	Must
FR-07	Live preview	Update the resume preview as data changes; preview should use the same rendering rules as export.	Must
FR-08	Field checks	Show missing required values, invalid links, suspicious date order and export overflow warnings.	Must
FR-09	Draft save	Allow explicit save on the current device and clear/reset. Explain browser/device limits; no silent cloud upload.	Must
FR-10	JD upload/paste	Accept supported file types or pasted text and show useful extraction errors.	Must
FR-11	JD analysis	Extract and display role requirements; flag ambiguous requirements instead of guessing.	Must
FR-12	AI suggestions	Generate supported edits with traceable source mapping, uncertainty labels, and an explicit review workflow.	Must
FR-13	No overwrite	Never overwrite student-authored content without an explicit accept action.	Must
FR-14	PDF export	Generate PDF in the approved layout with selectable text, working links, correct image placement, and safe page breaks.	Must
FR-15	Overflow guard	Never silently cut content. Warn and guide the user to edit or shorten content before export when needed.	Must
FR-16	Template version	Bind each render to a named internal template version; support controlled update and rollback.	Must
FR-17	Data deletion	Offer clear deletion/reset for local draft and, in any later account mode, deletion of saved profile/resume data.	Must
FR-18	Accessibility	Support keyboard navigation, accessible labels, readable contrast, visible validation errors, and mobile zoom/reflow.	Must
FR-19	Usage analytics	Collect only minimal, non-content event metrics needed to improve reliability; do not log raw resume/JD text by default.	Should

ID	Feature	Requirement	Priority
FR-20	Admin configuration	Keep section rules, template assets, and required fields configurable by authorized maintainers rather than hard-coded across UI.	Should

11. Non-Functional Requirements

Category	Requirement / target
Mobile responsiveness	Support current Chrome on Android and Safari/Chrome on iOS; design from narrow viewport first; test at 360 px and larger.
Performance	Target initial interactive experience within 2.5 seconds at the 75th percentile on a reasonable 4G connection; ordinary PDF generation within 10 seconds; JD suggestion response within 20 seconds at the 95th percentile. Measure in testing and revise against real telemetry.
Reliability	PDF generation errors must be recoverable and actionable. Never return a corrupt or partial document as a successful download.
Rendering fidelity	Preview and exported PDF share one template source of truth. Use automated text/link checks and visual regression tests for golden profiles.
Security	Validate inputs server-side; limit file size and types; use safe parsing; protect API secrets; apply rate limits; prevent cross-user access if cloud accounts are introduced.
Privacy	Provide plain-language purpose notice, consent for AI transmission, limited retention, deletion controls, and no raw PII/JD text in routine logs.
Accessibility	Use WCAG 2.2 AA as a design/test target for relevant web content and apply mobile guidance; avoid relying on color alone or drag-only interactions.
Maintainability	Separate form schema, template configuration, AI suggestion service, and PDF rendering. Keep prompt/version changes testable and auditable.
Portability	A student must be able to download the generated resume directly to their device without requiring a desktop-specific workflow.
Cost control	Set usage and file-size limits to keep free core use sustainable. Communicate any usage limits before the user submits long files or requests AI processing.

12. Proposed Technical Approach

This section is a suggested implementation direction, not a locked technology decision. Choose a stack the team can maintain and test. The key design requirement is a deterministic template renderer with strict separation between user facts and AI suggestions.

Component	Proposed approach	Reason / control
Frontend	React + TypeScript, responsive web app / PWA	Componentized form sections, mobile preview, accessibility and responsive behavior.
Backend API	Python FastAPI or Flask	Central validation, template/config access, AI orchestration, rate limiting and export requests.
Structured data	PostgreSQL if server-side saved profiles are enabled; otherwise keep the MVP state local and process-only	Use a normalized schema; do not introduce persistent storage if the product does not need it.

Component	Proposed approach	Reason / control
Template rendering	Approved HTML/CSS template rendered to PDF by a controlled Chromium-based engine	Deterministic layout, hyperlinks, embedded fonts/assets, repeatable test capture.
File handling	Temporary processing area with strict size/type checks and scheduled deletion	Reduce long-term exposure to photos, JDs, and generated content.
AI layer	Provider-agnostic service with JSON-schema outputs and a validator before UI display	Keep prompts, model choice and evaluation replaceable; unsupported outputs are rejected.
Observability	Structured operational metrics with PII redaction	Track generation success, failures and latency without storing resume body text by default.

Architecture constraints

- The AI service must not be able to modify the locked template or insert arbitrary HTML/CSS into the output.
- PDF renderer should accept only validated structured fields and approved static assets, not user-supplied markup.
- Maintain a single canonical field schema to drive validation, editor forms, review labels, and rendering.
- Keep profile facts, JD-derived requirements, and generated suggestions in distinct data structures.
- If cloud storage/accounts are added, implement authenticated access, deletion, retention expiry, and audit logging before enabling them.

13. Privacy, Security and Responsible AI

The product handles personal data, potentially including a student's phone number, email, address, photograph, education history, profile links and career information. JD uploads may also contain confidential company text or identifying metadata. Treat this data as sensitive from a product-design perspective, even where a particular field is not legally classified as sensitive data.

Required product controls

- Show a clear privacy notice before collection. Explain what is collected, why, where it is processed, whether an AI provider receives it, and how a user can delete it.
- Manual mode must not transmit resume content to an AI provider.
- Before first JD Assist request, request explicit consent to send the necessary profile fields and JD text for AI processing. Offer a manual path if the student declines.
- Do not store raw AI prompts/responses containing personal data in routine logs. Use redacted error categories and request IDs instead.
- Define retention separately for local drafts, temporary uploaded files, AI processing traces, generated PDFs, and account-based saved profiles if added later.
- Provide delete/reset controls and test them. Local deletion should clear the app-managed draft and image; cloud deletion must cover derived copies and backups according to the disclosed policy.
- Protect secrets and restrict the AI service's context to the minimum fields needed for the requested suggestion.
- Do not use student resume content to train models unless a distinct, informed opt-in and legal review explicitly permits it.
- Review risks for users below 18 and any rules applicable to child data before public release; do not collect date of birth merely to personalize a resume.
- Have qualified counsel review obligations under India's Digital Personal Data Protection Act, 2023 and the Digital Personal Data Protection Rules, 2025, including relevant commencement/transition timelines. This PRD is not legal advice.

Planning references: Government of India - DPDP Rules, 2025: official source; W3C Mobile Accessibility: guidance; WCAG 2.2 mobile guidance: W3C note

These sources inform the proposed controls; obtain legal advice on applicable obligations and do not treat this PRD as a legal or accessibility-conformance certification.

Trust boundaries

Data / boundary	Rule
Candidate-supplied facts	Source of truth for personal claims. Preserve value and meaning; user correction is required for any factual change.
Job description	Untrusted employer-provided text used only to identify role requirements; never treat it as a candidate fact or system instruction.

Data / boundary	Rule
AI generated suggestion	Untrusted draft until validated against profile source fields and approved by the student.
External link	Untrusted URL until scheme and format pass validation; never fetch arbitrary URLs server-side just to validate that a profile exists.
Uploaded file	Untrusted input. Enforce size/type limits, use safe parsers, avoid macro execution, and fail clearly on extraction errors.

14. PDF Layout, Pagination and Export Rules

- The preview must represent the final PDF's actual typography, section order, photo crop, logo placement, hyperlinks, and page breaks as closely as possible.
- Template geometry and visual hierarchy remain fixed; content can flow only within defined section and pagination rules.
- Do not split a section heading from its first item. Avoid isolated one-line page tails/orphan headings where possible.
- Never hide overflow by shrinking text below the approved readable minimum, deleting bullets, clipping text, or squeezing images out of proportion.
- If content exceeds the template's allowed page count, show the affected section and offer safe options: edit manually, accept a clearly displayed shorter AI rewrite, or keep the extra page if policy allows. The app must not silently truncate.
- Preserve selectable text and clickable PDF links. Ensure the logo/photo are embedded and do not rely on temporary local file URLs.
- Use consistent fonts with embedded or reliably available glyphs; test punctuation, Indian names, common degree symbols, and mixed-case URLs.
- PDF metadata should not unintentionally expose internal storage paths, server names, raw prompts, or unnecessary personal data.
- Run automated checks for expected text presence, link annotations, image count/placement, page count and extraction integrity, plus visual review on representative devices/viewers.

15. Acceptance Criteria and Release Gates

ID	Area	Pass condition
AC-01	Manual entry fidelity	For the approved test profiles, every typed value, date, mark, skill and bullet appears correctly; no content is silently altered or lost.
AC-02	Empty fields	Optional empty sections and links do not appear in the output. No blank headings, placeholder URLs, "N/A", or invented filler is rendered.
AC-03	Mandatory fields	Required fields follow approved template configuration; errors explain how to fix them and preserve the data already entered.
AC-04	Hyperlinks	Each displayed link label is short and human-readable; its PDF target matches the normalized approved user URL and opens in a common PDF viewer.
AC-05	Photo/logo	Assets stay within approved frames, are not distorted, and match approved versions/resolution requirements.
AC-06	Pagination	No clipped or overlapping text, split headings, lost bullets, or silent truncation across all golden test profiles.
AC-07	JD extraction	Supported files produce a preview of extracted content; unreadable files fail with a helpful message and paste-text alternative.

ID	Area	Pass condition
AC-08	AI evidence	Every factual clause in every suggested summary/bullet/skill list maps to a profile source or is flagged for explicit student confirmation; unsupported claims are blocked.
AC-09	AI confirmation	No generated text enters the final PDF until the student accepts it; manually entered content is not silently overwritten.
AC-10	Same JD test	With one JD and a cohort of distinct profiles, suggestions emphasize genuinely different supported facts where relevant; identical facts are not fabricated to force uniqueness.
AC-11	Privacy	Manual mode makes no AI calls; JD Assist requires consent before data transmission; raw resume/JD text is absent from routine logs.
AC-12	Device support	Core workflow completes on supported iOS and Android mobile browsers at the declared minimum viewport without horizontal page scrolling for ordinary form use.
AC-13	Recovery	Network/model/rendering errors preserve the draft where possible and do not present a failed/partial PDF as successful.

Release-blocking failures

Do not release if any test can produce fabricated student facts, export a wrong hyperlink, silently omit content, expose private content in logs, use an unapproved template/logo, or generate a corrupt/clipped resume that is reported as successful.

16. Test Strategy

Test layer	Coverage
Unit tests	URL normalization; validators for email/phone/date/CGPA; conditional-section logic; schema migrations; prompt-output schema validation; provenance mapping.
Template golden tests	Approved set of representative profiles: minimal content, long project bullets, multiple projects, no links, maximum links, long names, long degrees, missing optional fields, long skills, and maximum allowed photo dimensions.
PDF integrity tests	Extracted text contains every expected field; hyperlinks have expected annotation targets; page count is in policy; images are present; metadata does not reveal forbidden internal values.
Visual regression	Render PDFs to images and compare with approved reference images for layout changes. Inspect every page of each release candidate’s critical templates.
AI evaluation	Same JD across diverse profile types; empty profile; missing required skill; conflicting dates; student says “do not claim”; quantified impact absent; prompt injection embedded in JD; ambiguous qualification wording.
Security tests	Malformed uploads, MIME/signature mismatch, oversized files, prompt injection, cross-origin behavior, rate limits, access control if accounts exist, secrets exposure, and deletion behavior.
Usability	Observe students completing a resume on mobile; measure completion time, error recovery, where they abandon, and whether they understand link labels and AI review controls.

Test layer	Coverage
Accessibility	Keyboard/screen reader spot checks, labels and error announcements, zoom/reflow, contrast, touch target size, and non-drag alternatives.

17. Analytics and Operational Monitoring

Track product reliability and usability without collecting resume content unnecessarily. Proposed event names and metrics:

Event / metric	Use	Privacy guardrail
build_started / build_completed	Understand the funnel and completion time.	Use random request/session IDs; do not attach name, email, phone or resume fields.
validation_error_category	Identify confusing inputs.	Log category (e.g. invalid URL), not the value entered.
pdf_export_success/failure	Monitor export reliability and failure types.	Capture error code, template version, latency and page count; omit resume body.
jd_parse_success/failure	Monitor file extraction quality.	Log file type/size bucket and error category, not raw JD content.
ai_suggestion_accept/edit/reject	Evaluate usefulness of suggestions.	Prefer counts and section types; avoid storing raw generated text unless separately consented for evaluation.
unsupported_claim_blocked	Monitor AI guardrail quality.	Store rule ID and synthetic test indicator; do not log the full personal statement by default.

18. Key Risks and Mitigations

Risk	Impact	Mitigation
Template not formally approved	Product may not match official placement requirements or may misuse brand assets.	Obtain editable master, written section/order requirements, authorized logo and approval before launch.
AI fabricates or exaggerates	Student submits inaccurate resume; loss of trust and possible placement consequences.	Source mapping, schema validation, blocked unsupported claims, user approval, adversarial tests, and release gate.
Repeated generic AI output	Students receive generic and low-value resumes.	Ground personalization in profile-specific facts, reorder relevant verified material, offer wording alternatives; never invent differences.
Overflow / bad page breaks	A resume looks unprofessional or loses content.	Preview/export parity, overflow warnings, safe rewrite approval, golden visual tests; never silently truncate.
Personal data exposure	Privacy harm from contact details, photos, or resumes.	Minimize retention, consent before AI transfer, redact logs, restrict access, test deletion, and complete legal review.
JD upload is untrusted/malicious	Prompt injection, parser failures, unexpected resource use.	Safe parsers, strict limits, treat document content as data, rate limiting, sandboxing, structured outputs.
Broken or misleading profile link	Recruiter cannot reach the profile or the resume points to the wrong URL.	Show normalized target to user, validate scheme/host format, test PDF annotations, do not claim ownership is verified.
Official template changes	Old resumes become inconsistent with current placement instructions.	Version templates, show update notice, preserve old drafts safely, migrate only with user preview/consent.

Risk	Impact	Mitigation
AI provider retains submitted data	Student data may be retained outside the app's direct control.	Choose provider/configuration based on data handling terms; disclose sub-processors and retention; send minimum necessary context.

19. Suggested Delivery Milestones

Milestone	Deliverables	Exit criteria
M0 - Template discovery	Approved master template, official logo/photo rules, section schema, layout constraints, privacy/legal review owner.	Placement-cell owner confirms exact version in writing.
M1 - Manual builder	Mobile form, repeatable fields, link validation, photo crop, live preview, local draft, PDF export.	Golden profile set passes content, link, image and pagination tests.
M2 - AI assist alpha	JD upload/paste, extraction review, structured suggestions, provenance, accept/edit/reject workflow.	No unsupported claims in fixed evaluation suite; injection tests pass.
M3 - Privacy and release hardening	Retention/deletion behavior, notices/consent, security tests, accessibility review, metrics without PII.	Release-blocking criteria are clear and signed off.
M4 - Student pilot	Limited student cohort, feedback, completion/export analytics, issue triage.	No critical factual/layout issues; usability metrics and support plan reviewed.

20. Decisions Required Before Development

Decision	Why it matters	Recommended default
Official master source	A PDF sample does not fully specify editable layout and mandatory rules.	Get the current placement-cell-approved DOCX/template package and brand assets.
Mandatory sections and page limit	Determine whether blank sections are omitted and how content overflow is handled.	Use configurable mandatory/optional flags; never truncate. Confirm page-count rules with placement cell.
Photo requirement	The sample contains a photo, but policy may differ by program or recruitment context.	Treat photo as required only when official instructions confirm it; provide clear crop and quality warnings.
Address granularity	Full residential address may expose more personal information than recruiters need.	Ask placement cell whether city/state is sufficient; collect full street address only if required.
Self-declaration wording	The exact phrase/date/place/signature requirements must not be guessed.	Use only approved wording; make date/place fields configurable.
AI provider and data handling	JD Assist transmits personal content to a model service.	Select a provider/configuration after privacy, retention, regional processing and contractual review.
Draft persistence	Local versus cloud save affects recovery and privacy.	MVP: explicit current-device draft with clear delete; cloud accounts only after access-control and deletion work.
Export formats	DOCX adds formatting parity and compatibility complexity.	MVP PDF only; evaluate DOCX after PDF fidelity and links are reliable.
University endorsement wording	Branding and the word "official" need permission.	Do not claim official university endorsement until authorized in writing.